

CARBONEASE

A Carbon Credit Marketplace

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1. Abstract

CarbonEase is a revolutionary platform aimed at simplifying carbon credit transactions through a user-friendly, transparent, and automated system. Carbon emissions are a growing concern globally, and carbon credits serve as a mechanism to offset these emissions by promoting green practices. Our application leverages a hybrid technology stack involving MERN (MongoDB, Express.js, React.js, Node.js) for the web interface and Flutter for the mobile app, making the platform both scalable and cross-platform.

The application supports functionalities such as real-time carbon emission tracking, automated credit matching, GPS-enabled activity verification, and secure trading mechanisms. The platform intends to bridge the gap between carbon emitters and those with excess carbon credits, allowing for seamless and traceable exchanges that contribute to environmental sustainability.

2. Introduction

In recent years, awareness around climate change has surged, leading to increased regulations and incentives for industries to reduce carbon footprints. Carbon credits have emerged as an essential market instrument, encouraging entities to adopt eco-friendly technologies. However, navigating the carbon credit market is often complicated due to technical, regulatory, and accessibility barriers.

CarbonEase aims to eliminate these barriers by offering a platform where users can easily track, buy, and sell carbon credits based on their carbon usage or green contributions. The solution not only empowers businesses to take control of their carbon output but also encourages widespread participation in the sustainability movement by making the process efficient, digital, and transparent.

3. Related Work

Numerous platforms such as Gold Standard, ClimateTrade, and Verra have laid the foundation for global carbon markets. These platforms verify carbon offset projects and issue credits. However, these systems often require substantial documentation, are tailored for large enterprises, and lack real-time feedback or mobile access.

Our research also evaluated initiatives like Klima DAO and Toucan Protocol, which focus on tokenized carbon credits. While innovative, these platforms demand blockchain familiarity and lack regional adaptability. CarbonEase builds upon these insights by simplifying entry, integrating essential features into a mobile-first solution, and using conventional APIs for verifiable carbon footprint calculations, enabling faster onboarding and broader adoption.

4. Tech Stack

Frontend: React.js for Web, Flutter for Mobile

Backend: Node.js with Express.js

Database: MongoDB

Hosting: AWS / Google Cloud Platform

APIs & Integrations:

Google Maps API for location tracking

Carbon Emission APIs for footprint calculations

Twilio for OTP verification

Firebase for push notifications and analytics

Tools: GitHub, Postman, Tailwind CSS, Vite

This technology stack was chosen for its high performance, scalability, and developer ecosystem. The use of component-based frameworks and real-time backend communication enables a smooth and responsive user experience.

5. Implementation

The CarbonEase platform features three user roles: Admin, Business, and Verifier.

User Registration & Login: JWT-based authentication provides secure access.

Carbon Calculator: Integrates emission data APIs to generate accurate estimates based on business activity.

Marketplace: Businesses can list or browse carbon credits for purchase.

Verification System: A verifier role ensures legitimacy through document and GPS verification.

Notifications: Real-time alerts keep users updated on transaction status, verification progress, and platform updates.

Mobile App: The Flutter application provides the same core functionality, adapted for mobile-first interactions.

The system ensures both buyer and seller confidence through transparency and accountability in carbon trading activities.

6. Results

Initial testing demonstrated that the platform effectively processes transactions and updates dashboards in real time.

Carbon footprint data was successfully calculated based on mock inputs from industries.

The marketplace interface allows users to find and filter credits easily.

Location-based verification worked accurately during prototype testing.

The app achieved positive reviews from beta users in terms of design and usability.

Stress testing for concurrent users was conducted using Postman, and the system remained responsive under load conditions.

7. Conclusion

CarbonEase delivers a much-needed, accessible, and efficient solution for carbon credit trading. By using a modern tech stack and intuitive design, the platform bridges gaps in awareness, access, and security.

The implementation of emission calculators, real-time market dynamics, and GPS-enabled verification make the platform future-ready. With enhancements like blockchain-based transparency and gamification for engagement, CarbonEase has the potential to scale and contribute significantly to the global sustainability movement.

8. References

Verra - <https://verra.org>

Gold Standard - <https://www.goldstandard.org>

ReactJS Documentation - <https://reactjs.org>

NodeJS Documentation - <https://nodejs.org>

MongoDB Documentation - <https://www.mongodb.com>

Flutter Documentation - <https://flutter.dev>





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